

Giving Computers the Ability to Learn from Data

Dr. Md. Shahidur Rahman
Professor, DoCSE, SUST
rahmanms@sust.edu

What is Machine Learning?

- **Machine learning** is the application and science of algorithms that make sense of data.
- **Machine learning** is the field of study that gives computers the ability to learn without being explicitly programmed” —Arthur L. Samuel, AI pioneer, 1959
- “A computer program is said to learn from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, improves with experience **E**.” —Tom Mitchell, Professor at Carnegie Mellon University

ML Applications

- Near-human-level image recognition/ classification
- Near-human-level speech transcription
- Near-human-level handwriting transcription
- Dramatically improved text-to-speech conversion
- Digital assistants such as Google Assistant and Amazon Alexa
- Near-human-level autonomous driving
- Improved ad targeting, as used by Google, Baidu, or Bing
- Improved search results on the web
- Ability to answer natural language questions
- Medical Diagnostic - Cancer detection

Driving forces of machine learning

Three technical forces are driving advances in machine learning:

- Datasets and benchmarks
- Hardware
- Algorithmic advances

Data

If ML is the steam engine of this revolution, then data is its coal:

- User-generated image tags on Flickr, for instance, have been a treasure trove of data for computer vision.
 - YouTube videos,
 - Wikipedia
- ImageNet dataset, consisting of 1.4 million images that have been hand annotated with 1,000 image categories
- Bengali to English Translation Dataset (2.75 million sentence pairs)

Hardware

- The NVIDIA Titan RTX, a GPU that cost \$2,500 at the end of 2019, can deliver a peak of 16 teraFLOPS in single precision (16 trillion float32 operations per second).
 - That's about 500 times more computing power than the world's fastest supercomputer from 1990, the Intel Touchstone Delta.
- In 2020, the third iteration of the TPU card represents 420 teraFLOPS of computing power.
 - That's 10,000 times more than the Intel Touchstone Delta from 1990.
- Google revealed its Tensor Processing Unit (TPU) project

Algorithm

- Better *activation functions* for neural layers
- Better *optimization schemes*, such as *RMSProp* and *Adam*
- Transformer based Models
- ConvNets

Three types of ML

Supervised Learning

- > Labeled data
- > Direct feedback
- > Predict outcome/future

Unsupervised Learning

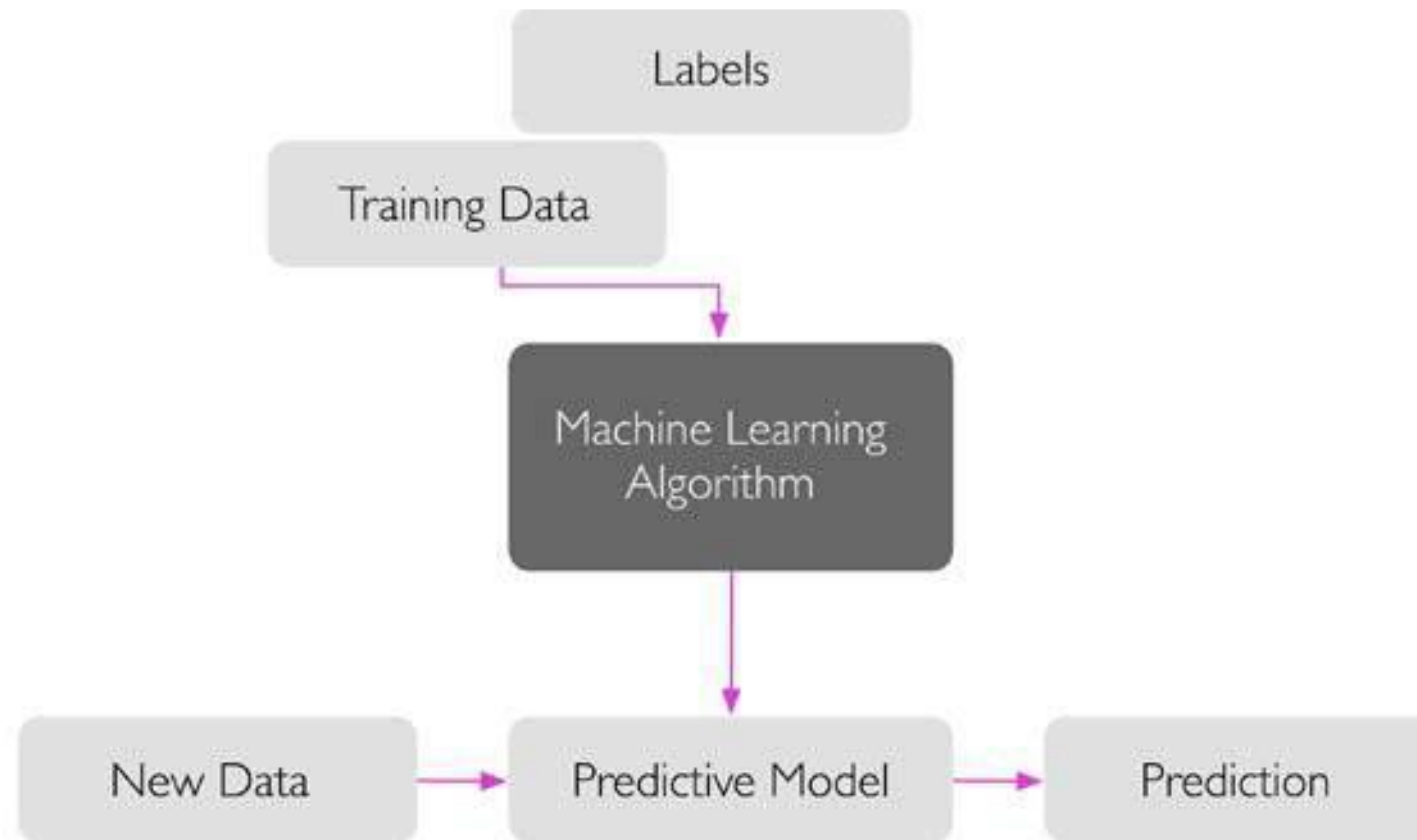
- > No labels
- > No feedback
- > Find hidden structure in data

Reinforcement Learning

- > Decision process
- > Reward system
- > Learn series of actions

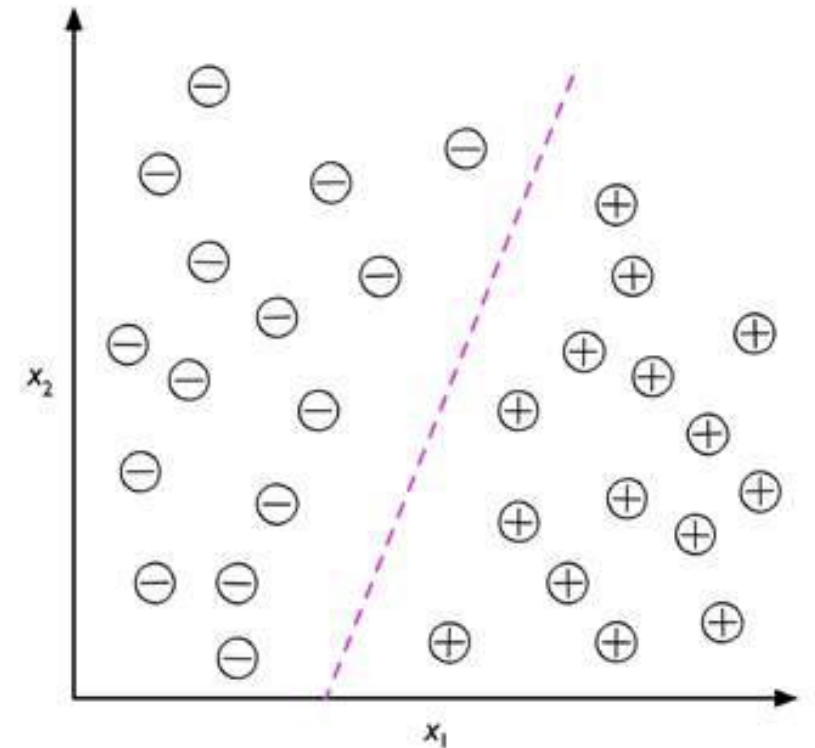
Making predictions with supervised learning

- Learn a model from **labeled training** data that allows us to make **predictions about unseen or future data**.
- For the email spam filtering example, we can train a model using a **supervised ML** algorithm on a corpus of labeled emails, which are correctly **marked as spam or non-spam**, to predict whether a **new email belongs to either of the two categories**.



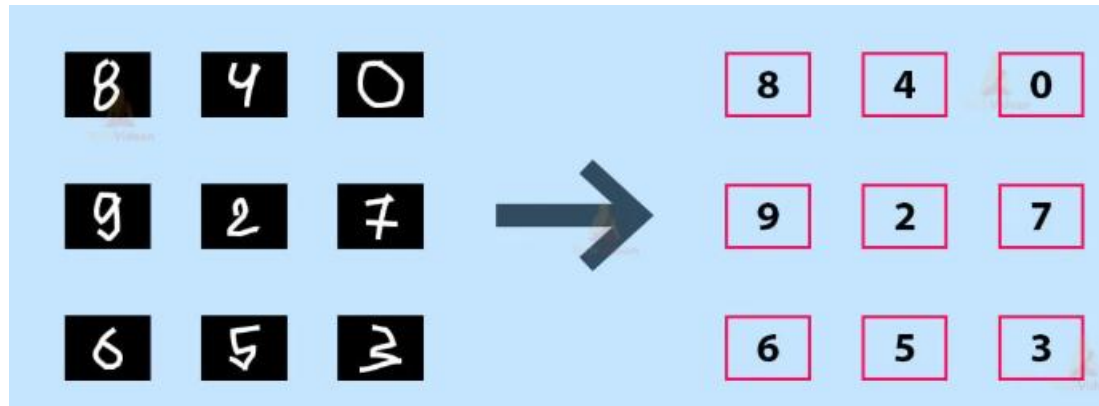
Classification for predicting class labels

- Supervised learning task with **discrete class labels**, such as in the previous email spam filtering example, is also called a **classification task**.
- A **binary classification** task given **30 training examples**; 15 training examples are labeled as the negative class (**minus signs**) and 15 training examples are labeled as the positive class (**plus signs**).
- A supervised ML algorithm learns a rule—**the decision boundary** (dashed line)



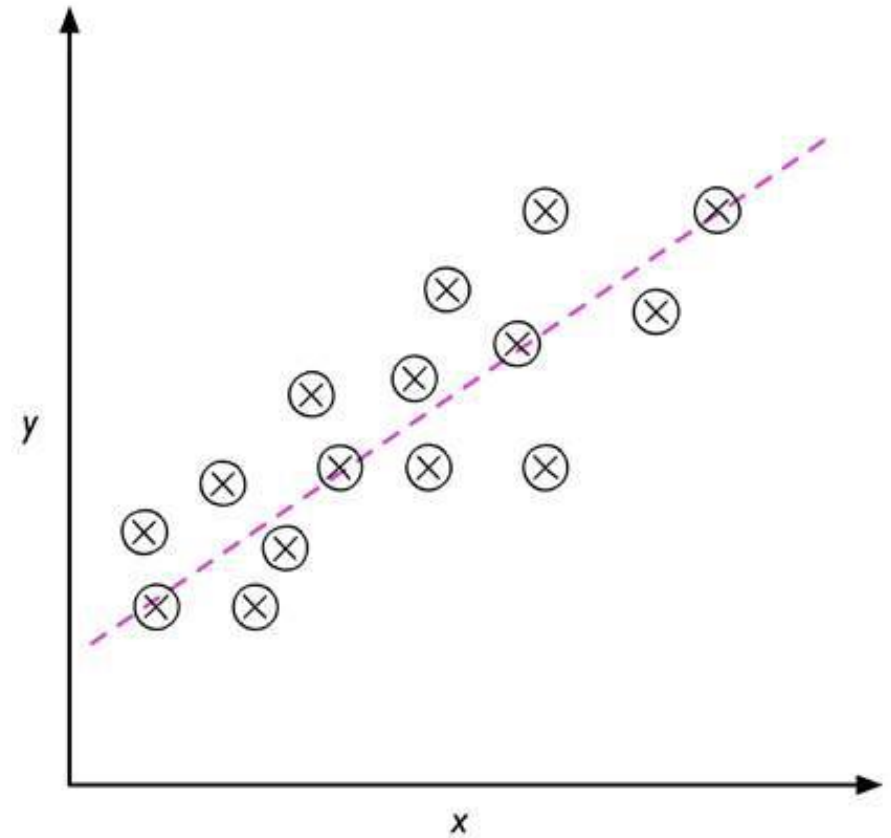
Multiclass classification

- A typical example of a multiclass classification task is **handwritten character recognition**.
- Also, **Digits recognition** between 0 and 9, is a multiclass classification



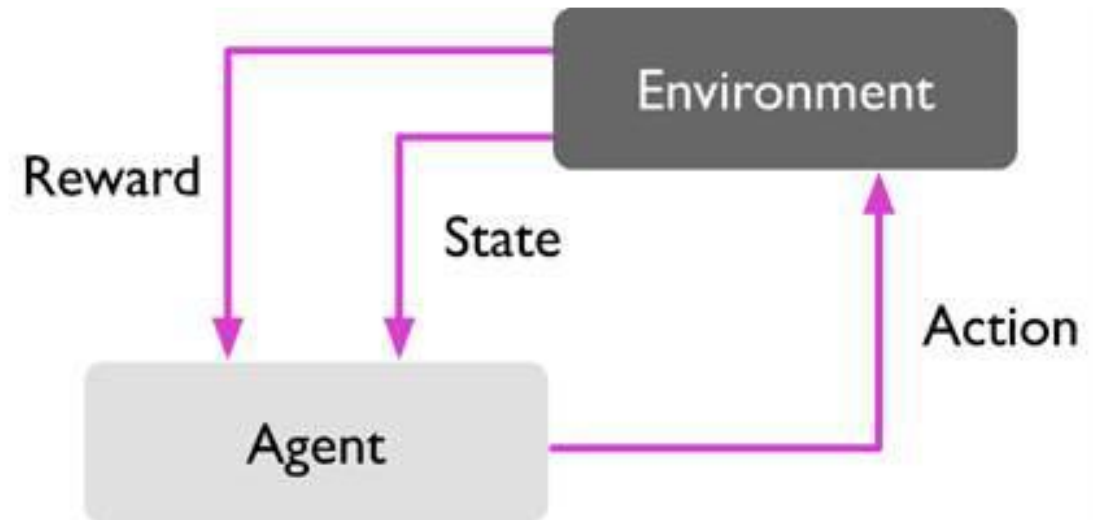
Regression for predicting continuous outcome

- Supervised learning where the outcome signal is a **continuous value** is **regression**
- We are given a number of predictor (**explanatory**) variables and a continuous response variable (**outcome**), and we try to find a **relationship** between those variables that allows us to predict an outcome.



Solving interactive problems with reinforcement learning

- In reinforcement learning, the goal is to develop a system (**agent**) that improves its performance based on **interactions with the environment**.
- Learns a **series of actions** that maximizes a **so-called reward** via an exploratory trial-and-error approach

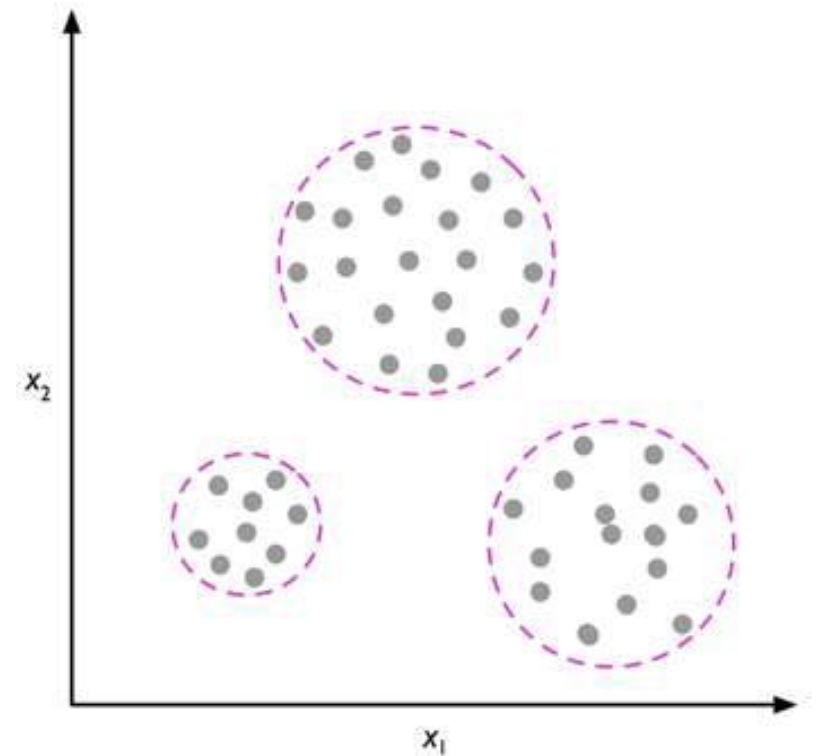


Discovering hidden structures with unsupervised learning

- In unsupervised learning deals with **unlabeled data** or data of **unknown structure**
- **Explore the structure of data** to extract **meaningful information** without the guidance of a known outcome variable

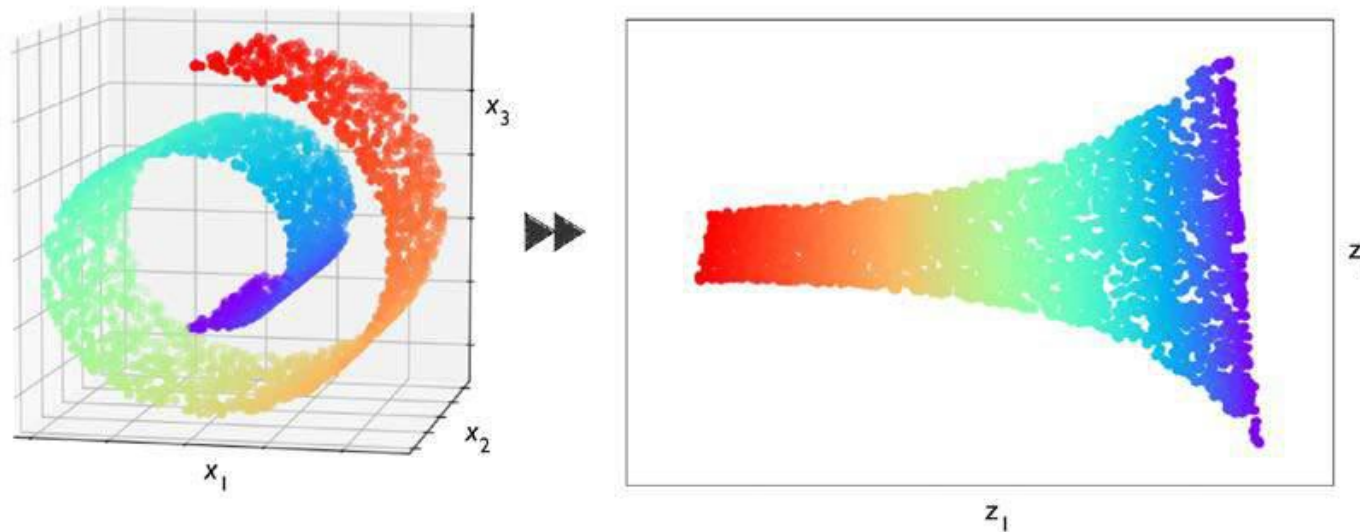
Finding subgroups with clustering

- **Clustering** is an exploratory data analysis technique that allows us to organize a pile of information into **meaningful subgroups (clusters)** without having any prior knowledge of their group memberships
- Example, three distinct groups based on the **similarity of their features, x_1 and x_2**



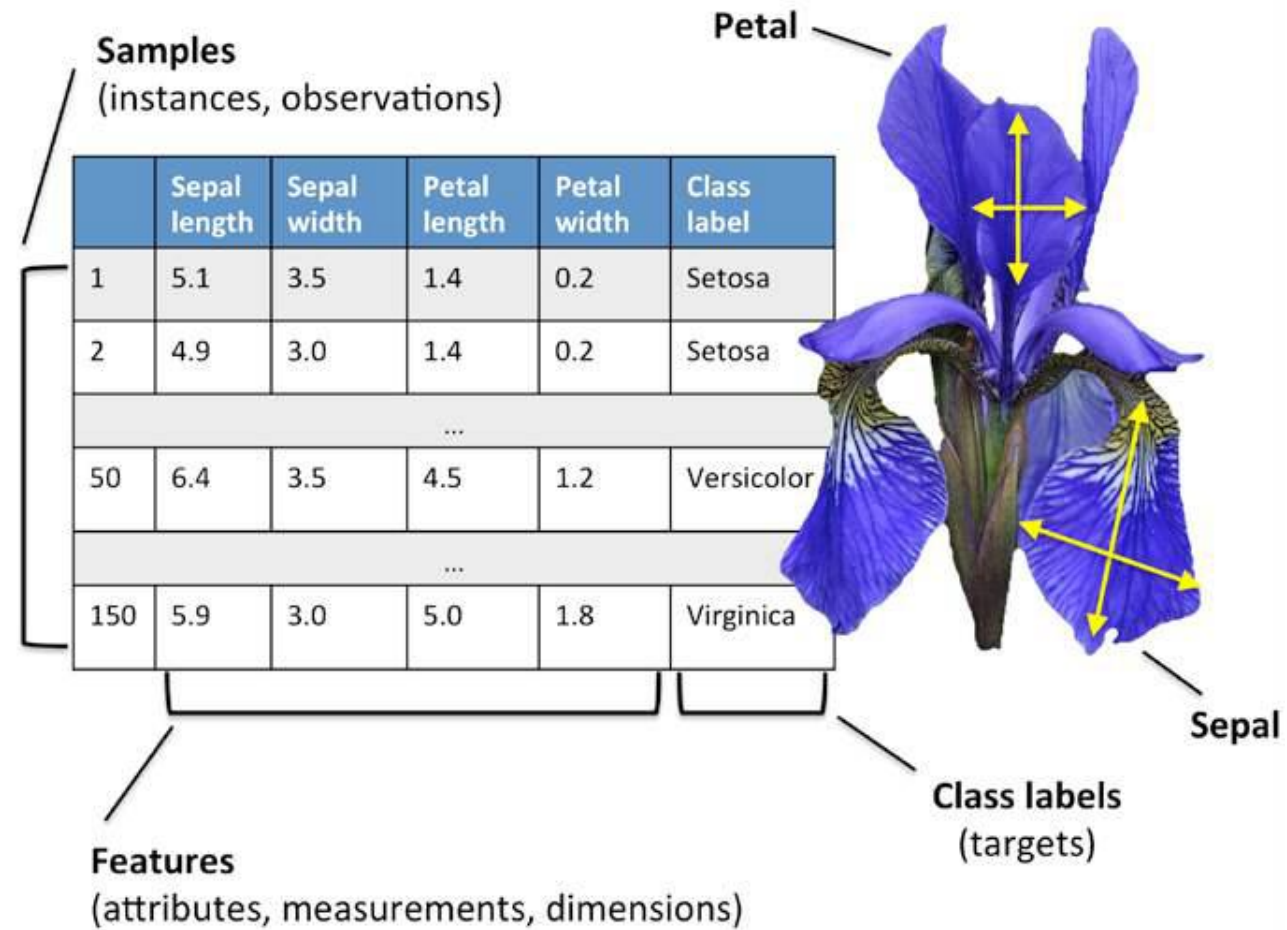
Dimensionality reduction for data compression

- A **high-dimensional feature** set can be projected onto **one-, two-, or three-dimensional** feature spaces in order to visualize it via 2D or 3D scatterplots or histograms.



- Example, Compress a 3D Swiss Roll onto a new 2D feature subspace:

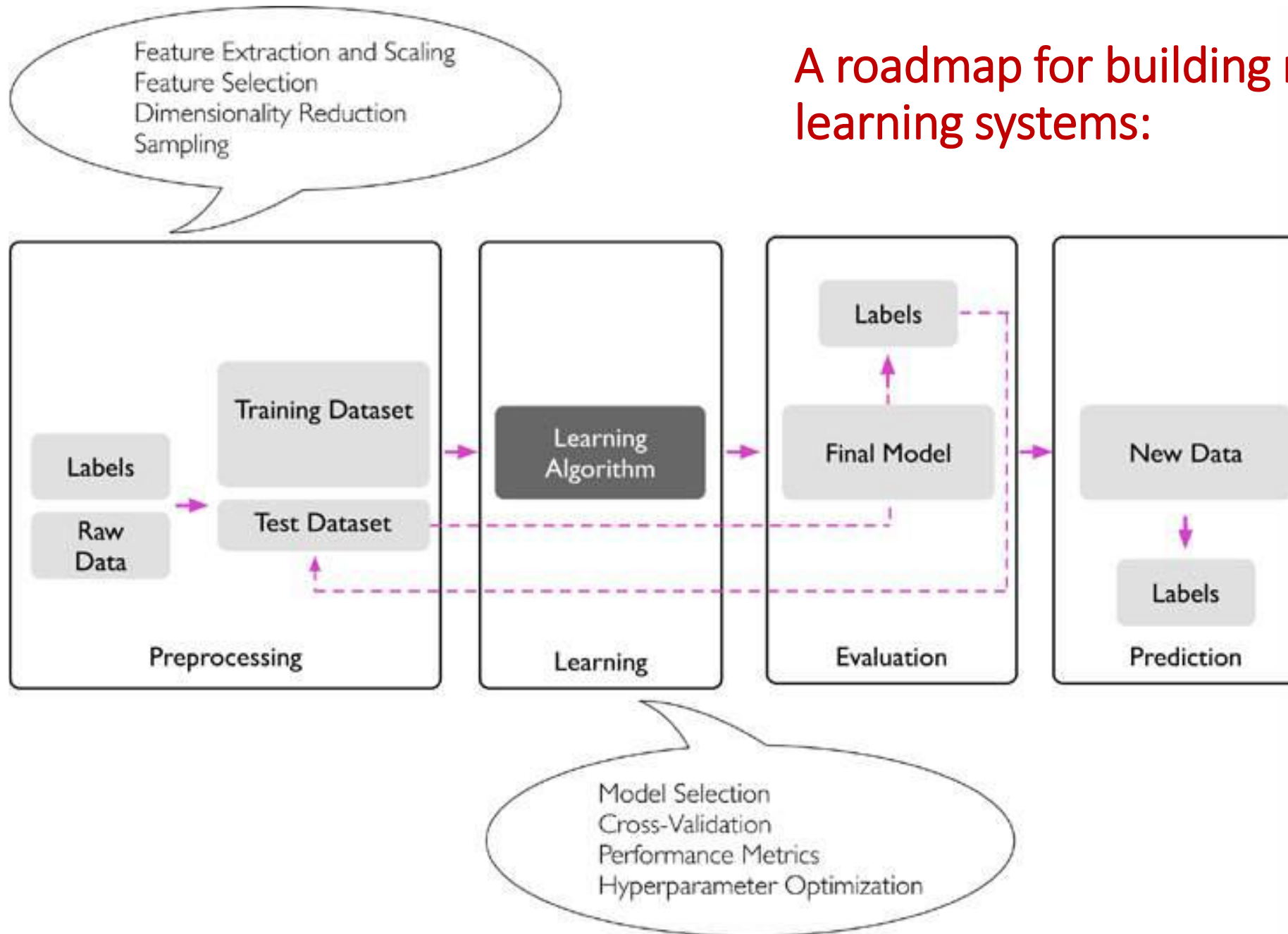
Basic terminology in ML



Basic terminology in ML

- **Training example:** A row in a table representing the dataset and synonymous with an observation, instance, or **sample**.
- **Training:** Model fitting, for parametric models similar to parameter estimation.
- **Feature**, abbrev. x : A column in a data table or data (design) matrix. Synonymous with variable, input, attribute, or covariate.
- **Target**, abbrev. y : Synonymous with outcome, output, response variable, dependent variable, (class) label, and ground truth.
- **Loss function:** Often used synonymously with a **cost function** or an **error function**.

A roadmap for building machine learning systems:



A roadmap for building machine learning systems

- **Preprocessing – getting data into shape**
 - Transforming the features in the range $[0, 1]$ or a standard normal distribution with **zero mean** and **unit variance**
- **Training and selecting a predictive model**
 - **Compare at least a handful of different algorithms** in order to train and select the best performing model.
- **Evaluating models and predicting unseen data instances**
 - After selecting a model that has been fitted on the training dataset, we can use the **test dataset to estimate how well it performs** on this unseen data